

RADAR LEVEL TRANSMITTER

TAGS: LT-01

Technical Information Micropilot FMR20B

Free-space radar
HART



Level measurement in liquids and bulk solids

Application

- Continuous, non-contact level measurement of liquids and bulk solids
- Degree of protection: IP66/68 / NEMA Type 4X/6P
- Maximum measuring range up to 30 m (98 ft)
- Process temperature: -40 to 80 °C (-40 to 176 °F)
- Process pressure: -1 to 3 bar (-14 to 43 psi)
- Accuracy: Up to ± 2 mm (0.08 in)
- International explosion protection certificates

Your benefits

- LED indicator for fast status detection
- Easy, guided commissioning with intuitive user interface
- Radar measuring device with *Bluetooth*® wireless technology and HART communication
- Easy, reliable and encrypted wireless remote access – ideal for difficult-to-reach installations, even in hazardous areas
- Commissioning, operation and maintenance via free iOS/Android app SmartBlue – saves time and reduces costs
- Flow measurement in open channels or weirs with totalizer

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About this document

Symbols

Safety symbols



This symbol alerts you to a dangerous situation. Failure to avoid this situation will result in serious or fatal injury.



This symbol alerts you to a potentially dangerous situation. Failure to avoid this situation can result in serious or fatal injury.



This symbol alerts you to a potentially dangerous situation. Failure to avoid this situation can result in minor or medium injury.



This symbol alerts you to a potentially harmful situation. Failure to avoid this situation can result in damage to the product or something in its vicinity.

Communication-specific symbols

Bluetooth®: 

Wireless data transmission between devices over a short distance via radio technology.

Symbols for certain types of information

Permitted: 

Procedures, processes or actions that are permitted.

Forbidden: 

Procedures, processes or actions that are forbidden.

Additional information: 

Reference to documentation: 

Reference to page: 

Series of steps: , , 

Result of an individual step: 

Symbols in graphics

Item numbers: 1, 2, 3 ...

Series of steps: , , 

Views: A, B, C, ...

List of abbreviations

PN

Nominal pressure

MWP

Maximum working pressure

The maximum working pressure is indicated on the nameplate.

ToF

Time of Flight

DTM

Device Type Manager

ϵ_r (Dk value)

Relative dielectric constant

Operating tool

The term "operating tool" is used in place of the following operating software:

- FieldCare / DeviceCare, for operation via HART communication, IO-Link communication and PC
- SmartBlue app for operation using an Android or iOS smartphone or tablet

PLC

Programmable logic controller (PLC)

Graphic conventions

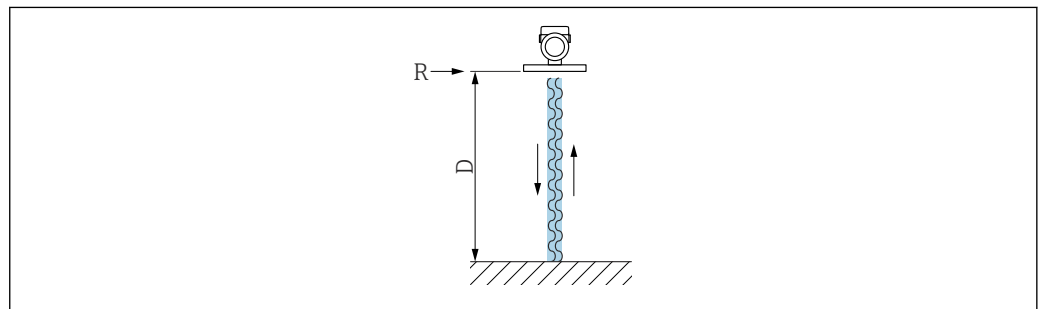


- Installation, explosion and electrical connection drawings are presented in simplified format
- Devices, assemblies, components and dimensional drawings are presented in reduced-line format
- Dimensional drawings are not to-scale representations; the dimensions indicated are rounded off to 2 decimal places
- Unless otherwise described, flanges are presented with sealing surface form EN 1092-1; ASME B16.5, RF.

Function and system design

Measuring principle

The Micropilot is a "downward-looking" measuring system, operating based on the frequency modulated continuous wave method (FMCW). The antenna emits an electromagnetic wave at a continuously varying frequency. This wave is reflected by the product and received again by the antenna.



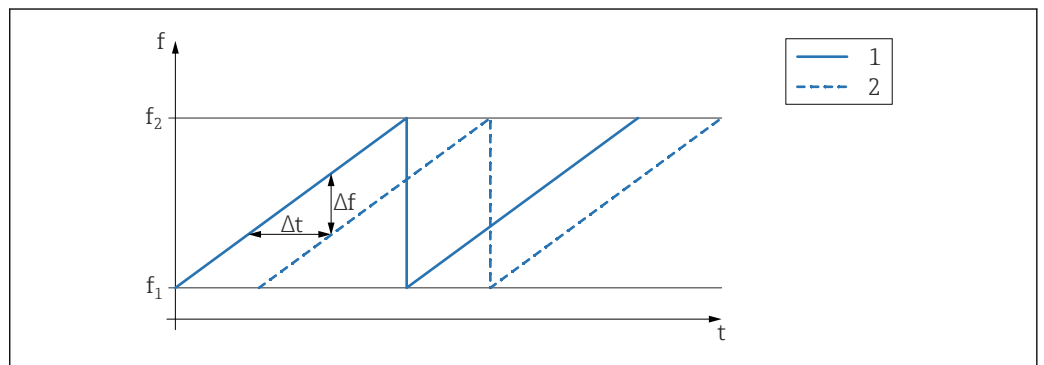
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1 FMCW principle: Transmission and reflection of the continuous wave

R Reference point of measurement

D Distance between reference point and product surface

The frequency of this wave is modulated in the form of a sawtooth signal between two limit frequencies f_1 and f_2 :



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2 FMCW principle: Result of frequency modulation

1 Transmitted signal

2 Received signal

This results in the following difference frequency at any time between the transmitted signal and the received signal:

$$\Delta f = k \Delta t$$

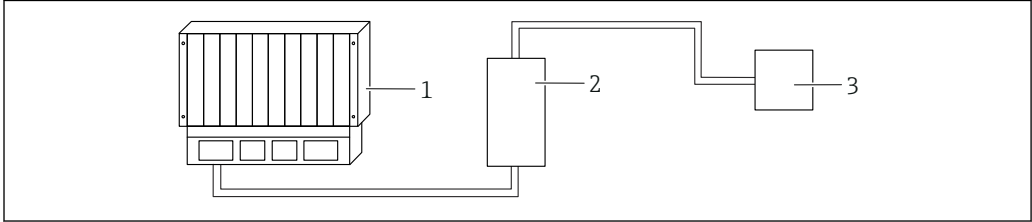
where Δt is the run time and k is the specified increase in frequency modulation.

Δt is given by the distance D between the reference point R and the product surface:

$$D = (c \Delta t) / 2$$

where c is the wave velocity.

In summary, D can be calculated from the measured difference frequency Δf . D is then used to determine the fill level or flow rate.

Measuring system	A complete measuring system comprises:  A0053220 1 PLC (programmable logic controller) 2 RMA42/RIA45 (if necessary) 3 Device
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Communication and data processing	■ 4 to 20 mA with superimposed digital communication protocol HART, 2-wire ■ Bluetooth® wireless technology (optional)
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Dependability	IT security The manufacturer warranty is valid only if the product is installed and used as described in the Operating Instructions. The product is equipped with security mechanisms to protect it against any inadvertent changes to the settings. IT security measures, which provide additional protection for the product and associated data transfer, must be implemented by the operators themselves in line with their security standards. Device-specific IT security The device offers specific functions to support protective measures by the operator. These functions can be configured by the user and guarantee greater in-operation safety if used correctly. The user role can be changed with an access code (applies to operation via Bluetooth® wireless technology or FieldCare, DeviceCare, Asset Management Tools (e.g. AMS, PDM)). <i>Access via Bluetooth® wireless technology</i> Secure signal transmission via Bluetooth® wireless technology uses an encryption method tested by the Fraunhofer Institute. ■ Without the SmartBlue app, the device is not visible via Bluetooth® wireless technology. ■ Only one point-to-point connection is established between the device and a smartphone or tablet. ■ The Bluetooth® interface can be disabled via SmartBlue or an operating tool via digital communication.
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Input

Measured variable	The measured variable is the distance between the reference point and the product surface. The level is calculated based on E, the empty distance entered.
Measuring range	The measuring range starts at the point where the beam hits the tank floor. Levels below this point cannot be measured, particularly in the case of spherical bases or conical outlets.

Maximum measuring range

The maximum measuring range depends on the antenna size.

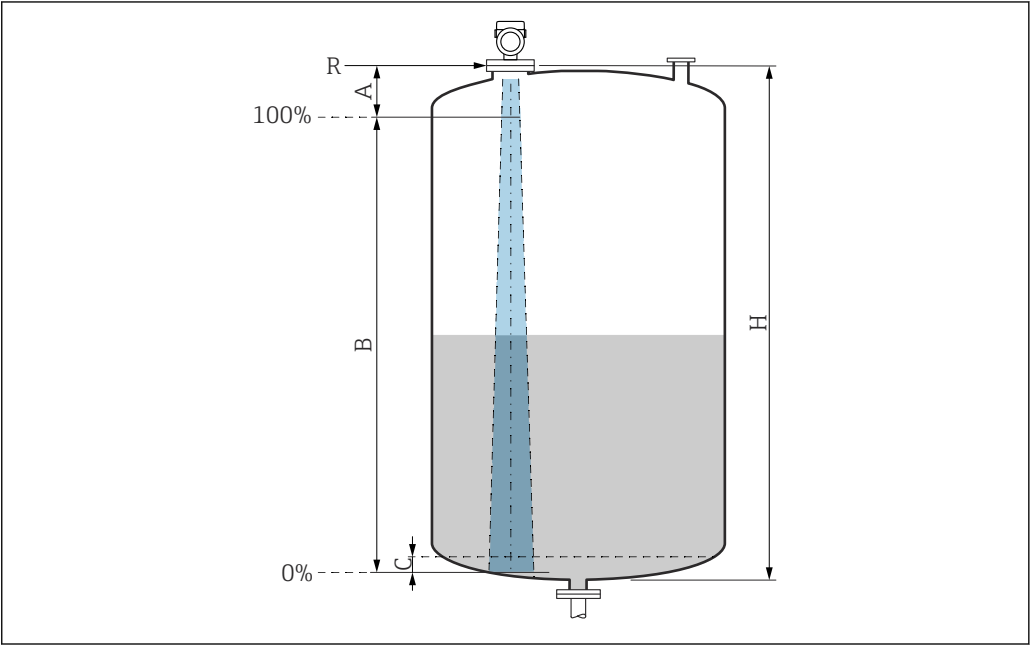
Antenna	Maximum measuring range
40 mm (1.5 in)	20 m (65.6 ft)
80 mm (3 in)	30 m (98.4 ft)

Usable measuring range

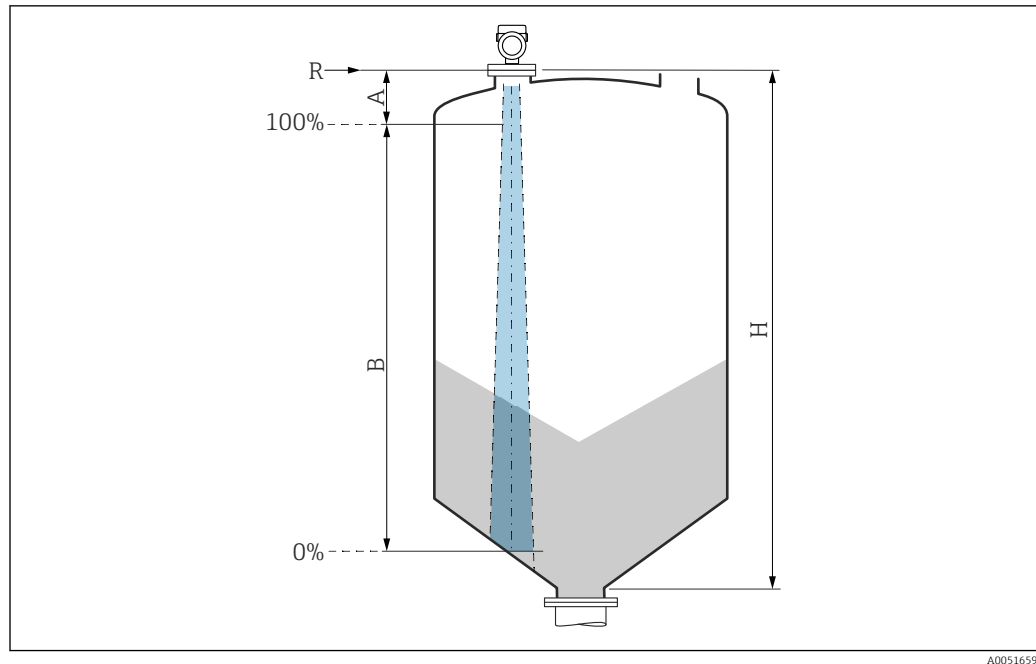
The usable measuring range depends on the antenna size, the medium's reflective properties, the installation position and any possible interference reflections.

In principle, measurement is possible up to the tip of the antenna.

Depending on the position of the product (angle of repose for solids) and to avoid any material damage from corrosive or aggressive media or deposit buildup on the antenna, the end of the measuring range should be selected 10 mm (0.4 in) before the tip of the antenna.



- A Antenna tip + 10 mm (0.4 in)
- B Usable measuring range
- C 50 to 80 mm (1.97 to 3.15 in); Medium $\epsilon_r \leq 2$
- H Vessel height
- R Reference point of the measurement, varies depending on the antenna system (see section on Mechanical construction)



- A** Antenna tip + 10 mm (0.4 in)
B Usable measuring range
H Vessel height
R Reference point of the measurement, varies depending on the antenna system (see section on Mechanical construction)

In the case of media with a low dielectric constant $\epsilon_r < 2$, the tank bottom may be visible through the medium when levels are very low (less than level C). Reduced accuracy must be expected in this range. If this is not acceptable, the zero point should be positioned at a distance C above the tank bottom for these applications (see figure).

The media groups and the possible measuring range are described as a function of the application and media group in the following section. If the relative permittivity of the medium is not known, to ensure a reliable measurement assume the medium belongs to group B.

Media groups

- **A** (ϵ_r 1.4 to 1.9)
Non-conductive liquids, e.g. liquefied gas
- **B** (ϵ_r 1.9 to 4)
Non-conductive liquids, e.g. gasoline, oil, toluene, etc.
- **C** (ϵ_r 4 to 10)
e.g. concentrated acid, organic solvents, ester, aniline, etc.
- **D** ($\epsilon_r > 10$)
Conductive liquids, aqueous solutions, diluted acids, bases and alcohol



For the relative permittivity values (ϵ_r values) of many media commonly used in industry, please refer to:

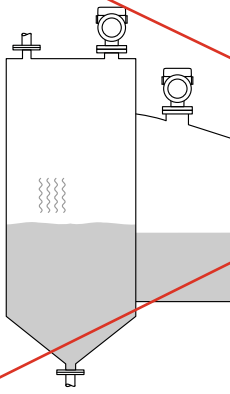
- Relative permittivity (ϵ_r value), Compendium CP01076F
- The Endress+Hauser "DC Values app" (available for Android and iOS)

Measurement in storage vessel

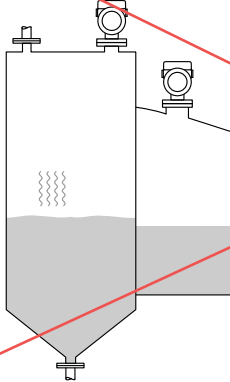
Storage vessel - measuring conditions

Calm medium surface (e.g. bottom filling, filling via immersion tube or rare filling from above)

40 mm (1.5 in) antenna in storage vessel

	Media group	Measuring range
	A (ϵ_r 1.4 to 1.9)	10 m (33 ft)
	B (ϵ_r 1.9 to 4)	20 m (65.6 ft)
	C (ϵ_r 4 to 10)	20 m (65.6 ft)
	D ($\epsilon_r > 10$)	20 m (65.6 ft)

80 mm (3 in) antenna in storage vessel

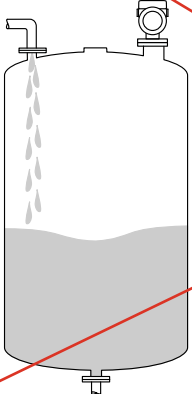
	Media group	Measuring range
	A (ϵ_r 1.4 to 1.9)	12 m (39 ft)
	B (ϵ_r 1.9 to 4)	23 m (75 ft)
	C (ϵ_r 4 to 10)	30 m (98 ft)
	D ($\epsilon_r > 10$)	30 m (98 ft)

Measurement in buffer vessel

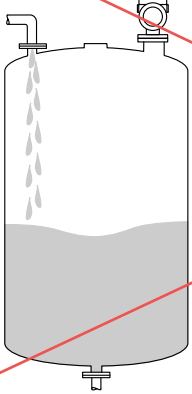
Buffer vessel - measuring conditions

Moving medium surface (e.g. permanent free filling from above, mixing jets)

40 mm (1.5 in) antenna in buffer vessel

	Media group	Measuring range
	A (ϵ_r 1.4 to 1.9)	7 m (23 ft)
	B (ϵ_r 1.9 to 4)	13 m (43 ft)
	C (ϵ_r 4 to 10)	20 m (65.6 ft)
	D ($\epsilon_r > 10$)	20 m (65.6 ft)

80 mm (3 in) antenna in buffer vessel

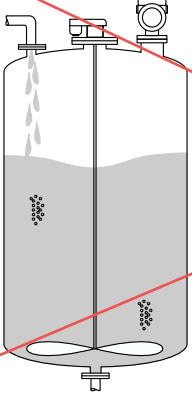
	Media group	Measuring range
	A (ϵ_r 1.4 to 1.9)	7.5 m (25 ft)
	B (ϵ_r 1.9 to 4)	15 m (49 ft)
	C (ϵ_r 4 to 10)	28 m (92 ft)
	D (ϵ_r >10)	30 m (98 ft)

Measurement in vessel with agitator

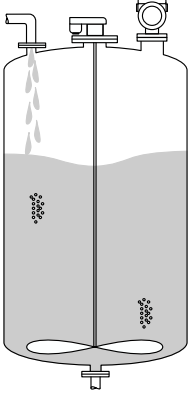
Vessel with agitator - measuring conditions

Turbulent medium surface (e.g. from filling from above, stirrers and baffles)

40 mm (1.5 in) antenna in vessel with agitator

	Media group	Measuring range
	A (ϵ_r 1.4 to 1.9)	4 m (13 ft)
	B (ϵ_r 1.9 to 4)	5 m (16.4 ft)
	C (ϵ_r 4 to 10)	13 m (43 ft)
	D (ϵ_r >10)	20 m (65.6 ft)

80 mm (3 in) antenna in vessel with agitator

	Media group	Measuring range
	A (ϵ_r 1.4 to 1.9)	4 m (13 ft)
	B (ϵ_r 1.9 to 4)	7 m (23 ft)
	C (ϵ_r 4 to 10)	15 m (49 ft)
	D (ϵ_r >10)	25 m (82 ft)

Operating frequency

approx. 80 GHz

Up to eight devices can be installed in a tank without the devices mutually influencing one another.

Transmission power

- Peak power: <1.5 mW
- Average output power: <70 µW

Output

Output signal

- 4 to 20 mA with superimposed digital communication protocol HART, 2-wire
- The current output offers a choice of three different operating modes:
 - 4 to 20.5 mA
 - NAMUR NE 43: 3.8 to 20.5 mA (factory setting)
 - US mode: 3.9 to 20.5 mA

Signal on alarm for devices with current output**Current output**

Signal on alarm in accordance with NAMUR recommendation NE 43.

- Max. alarm: can be set from 21.5 to 23 mA
- Min. alarm: < 3.6 mA (factory setting)

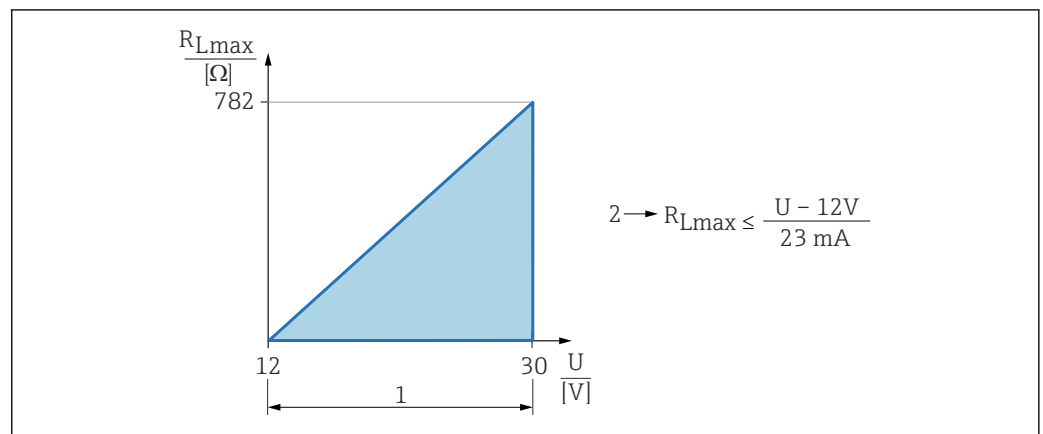
Operating tool via digital communication

Status signal (according to NAMUR Recommendation NE 107):

Plain text display

Load

In order to guarantee sufficient terminal voltage, a maximum load resistance R_L (including line resistance) must not be exceeded, depending on the supply voltage U of the supply unit.



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- 1 Power supply 12 to 30 V
 2 R_{Lmax} maximum load resistance
 U Supply voltage

If load is too great:

- Failure current is indicated and error message is displayed (indication: MIN alarm current)
- Periodic checking to establish if it is possible to quit fault state



Operation via handheld terminal or PC with operating program: take minimum communication resistor of 250 Ω into consideration.

Damping

Damping affects all continuous outputs.
 Factory setting: 0 s (can be set from 0 to 999 s)

Ex connection data

See the separate technical documentation (Safety Instructions (XA)) on www.endress.com/download.

Linearization

The linearization function of the device allows the conversion of the measured value into any unit of length, weight, flow or volume.

Pre-programmed linearization curves

Linearization tables for calculating the volume in the following vessels are pre-programmed into the device:

- Pyramid bottom
- Conical bottom
- Angled bottom
- Horizontal cylinder
- Sphere

Linearization tables for calculating the flow rate are pre-programmed into the device and include the following:

- Flumes
 - Khafagi Venturi flume
 - Venturi flume
 - Parshall flume
 - Palmer Bowlus flume
 - Trapezoidal flume (ISO 4359)
 - Rectangular flume (ISO 4359)
 - U-shaped flume (ISO 4359)
- Weirs
 - Trapezoidal weir
 - Rectangular broad-crested weir (ISO 3846)
 - Rectangular thin-plate weir (ISO 1438)
 - V-notch thin-plate weir (ISO 1438)
- Standard formula

Other linearization tables of up to 32 value pairs can be entered manually.



For further information on flow measurement over open channels and weirs, see SD03445F.

Totalizer	The device offers a totalizer which adds up the flow rate. The totalizer cannot be reset.
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Protocol-specific data	Manufacturer ID: 17(0x0011) Device type ID: 0x11DE Device revision: 2 HART specification: 7.6 DD version: 1
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Device description files (DTM, DD)

Information and files at:

- www.endress.com
On the product page for the device: Documents/Software → Device drivers
- www.fieldcommgroup.org

HART load:

Min. 250 Ω

The following measured values are assigned to the device variables at the factory:

Device variable	Measured value
Primary variable (PV) ¹⁾	Level linearized
Secondary variable (SV)	Distance
Tertiary variable (TV)	Absolute echo amplitude
Quaternary variable (QV)	Relative echo amplitude

1) The PV is always applied to the current output.

Choice of HART device variables

- Level linearized
- Distance
- Electronics temperature
- Sensor temperature
- Absolute echo amplitude
- Relative echo amplitude
- Area of incoupling
- Percent of range
- Loop current
- Flow
- Totalizer value
- Not used

Supported functions

- Burst mode
- Additional transmitter status
- Device locking

Wireless HART data

Minimum start-up voltage:
12 V

Start-up current:
< 3.6 mA

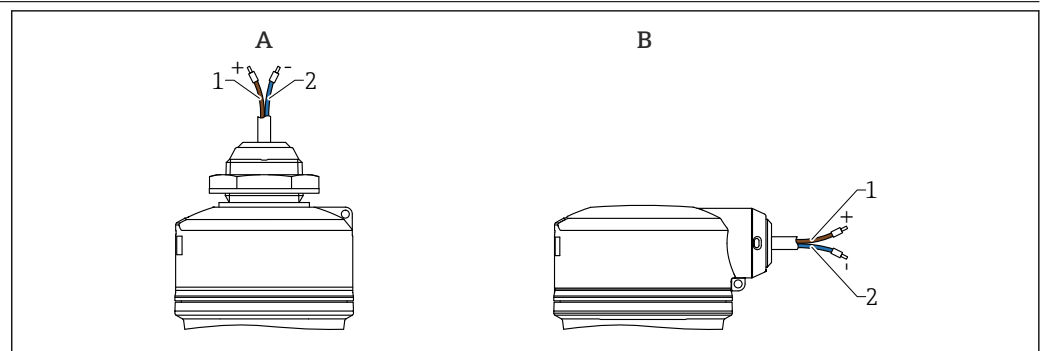
Starting time:
< 15 s

Minimum operating voltage:
12 V

Multidrop current:
4 mA

Time to establish connection:
< 30 s

Power supply

Cable assignment

A0055191

 3 *Cable assignment*

A Cable entry from above

B Side cable entry

1 Plus, brown wire

2 Minus, blue wire

Supply voltage

12 to 30 V DC on a DC power unit

 The power unit must be safety-approved (e.g. PELV, SELV, Class 2) and must comply with the relevant protocol specifications.

Protective circuits against reverse polarity, HF influences and overvoltage peaks are installed.

Power consumption

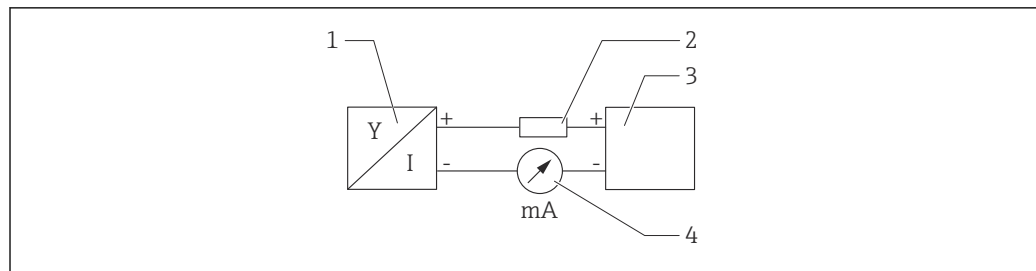
- Non-hazardous area: To meet device safety specifications according to the IEC 61010 standard, the installation must ensure that the maximum current is limited to 500 mA.
- Hazardous area: The maximum current is restricted to $I_i = 100$ mA by the transmitter power supply unit when the measuring instrument is used in an intrinsically safe circuit (Ex ia).

Potential equalization

No special measures for potential equalization are required.

Connecting the device**Function diagram of 4 to 20 mA HART**

Connection of the device with HART communication, power source and 4 to 20 mA indicator



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4 Function diagram of HART connection

- 1 Device with HART communication
- 2 HART resistor
- 3 Power supply
- 4 Multimeter or ammeter

i The HART communication resistor of 250 Ω in the signal line is always necessary in the case of a low-impedance power supply.

The voltage drop to be taken into account is:
Max. 6 V for 250 Ω communication resistor

Function diagram of HART device, connection with RIA15, display only without operation, without communication resistor

i The RIA15 remote display can be ordered together with the device.

b Alternatively available as an accessory, for details see Technical Information TI01043K and Operating Instructions BA01170K

Terminal assignment RIA15

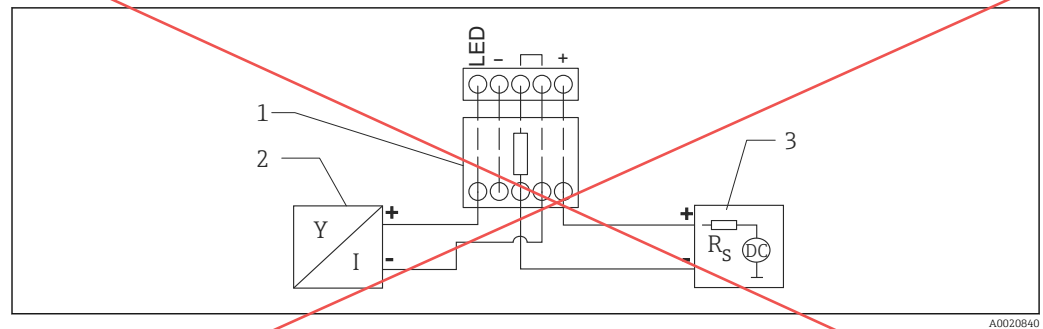
- +
Positive connection, current measurement
- -
Negative connection, current measurement (without backlighting)
- LED
Negative connection, current measurement (with backlighting)
- $\equiv$
Functional grounding: Terminal in housing

i The RIA15 process indicator is loop-powered and does not require any external power supply.

The voltage drop to be taken into account is:

- ≤ 1 V in the standard version with 4 to 20 mA communication
- ≤ 1.9 V with HART communication
- and an additional 2.9 V if display light is used

Connection of the HART communication resistor module, RIA15 with backlighting



8 Function diagram of HART device, RIA15 with light, HART communication resistor module

- 1 HART communication resistor module
 2 Device with HART communication
 3 Current supply

Cable specification

Unshielded cable, wire cross-section 0.5 mm²

- Resistant to UV and weather conditions as per ISO 4892-2
- Flame resistance according to IEC 60332-1-2

As per IEC 60079-11 section 9.4.4, the cable is designed for a tensile strength of 30 N (6.74 lbf) (over a period of 1 h).

The device is available in cable lengths of 5 m (16 ft), 10 m (32 ft), 15 m (49 ft), 20 m (65 ft), 30 m (98 ft) and 50 m (164 ft).

User-defined lengths up to total length of 300 m (980 ft) are possible in increments of one meter (order option "1") or one foot (order "2").

For devices with marine approval:

- Only available with a length of 10 m (32 ft) and "user-defined"
- Halogen-free as per IEC 60754-1
- No development of corrosive fire gases in accordance with IEC 60754-2
- Low flue gas density in accordance with IEC 61034-2

Overvoltage protection

The device satisfies the IEC/DIN EN 61326-1 product standard (Table 2 Industrial environment). Depending on the type of connection (DC power supply, input line, output line), different test levels are used to prevent transient overvoltages (IEC/DIN EN 61000-4-5 Surge) in accordance with IEC/DIN EN 61326-1: Test level for DC power supply lines and IO lines: 1 000 V wire to ground.

Devices for the "protection by enclosure" explosion protection are equipped with an integrated overvoltage protection system.

Overvoltage category

In accordance with IEC/DIN EN 61010-1, the device is intended for use in networks with overvoltage protection category II.

Performance characteristics

Reference operating conditions

- As per IEC 62828-1/IEC 62828-4
- Ambient temperature T_A = constant, in the range of +21 to +33 °C (+70 to +91 °F)
- Humidity ϕ = constant, in the range of: 5 to 80 % RH $\pm$ 5 %
- Atmospheric pressure p_U = constant, in the range of: 860 to 1 060 mbar (12.47 to 15.37 psi)
- Load with HART: 250 Ω
- Supply voltage: DC 24 V $\pm$ 3 V
- Reflector: metal plate with diameter $\geq$ 1 m (40 in)
- No major interference echoes inside the signal beam

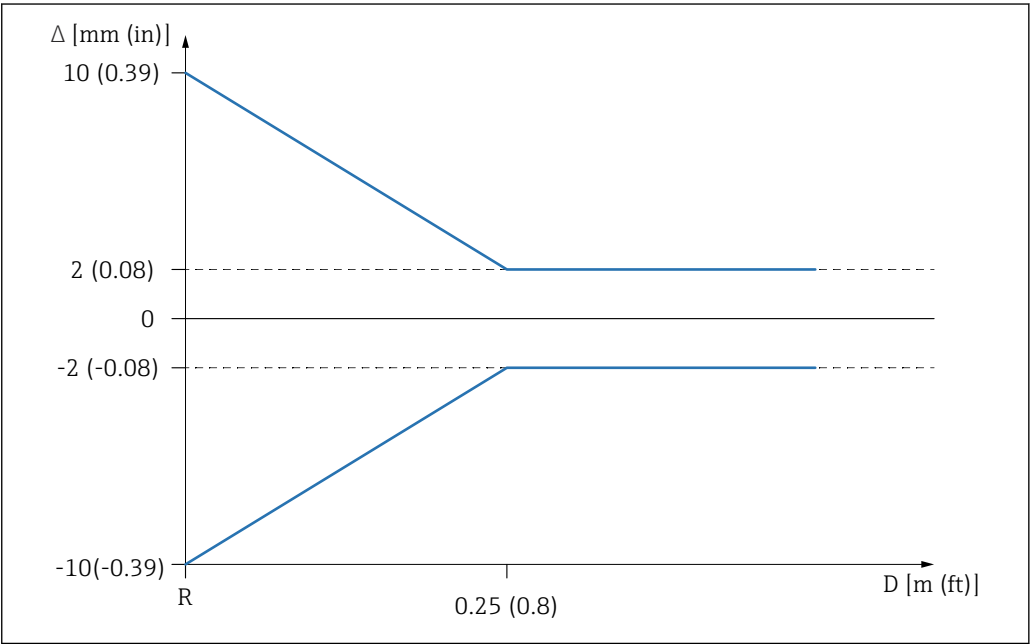
Response time

- HART: acyclic: min. 330 ms, typically 590 ms (depending on commands and number of preambles)
- HART: cyclic (burst): min. 160 ms, typically 350 ms (depending on commands and number of preambles)

Resolution	Current output: < 1 μ A Digital: 1 mm (0.04 in)
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Maximum measurement error	Reference accuracy Accuracy The accuracy is the sum of the non-linearity, non-repeatability and hysteresis. For liquids: ■ Measuring distance up to 0.25 m (0.82 ft): max. ± 10 mm (± 0.39 in) ■ Measuring distance > 0.25 m (0.82 ft): ± 2 mm (± 0.08 in) For solids: ■ Measuring distance up to 0.8 m (2.6 ft): max. ± 20 mm (± 0.79 in) ■ Measuring distance > 0.8 m (2.6 ft): ± 4 mm (± 0.16 in) Non-repeatability Non-repeatability is already included in the accuracy. ≤ 1 mm (0.04 in)  If conditions deviate from the reference operating conditions, the offset/zero point that results from the installation conditions can be up to ± 4 mm (± 0.16 in). This additional offset/zero point can be eliminated by entering a correction (Level correction parameter) during commissioning.
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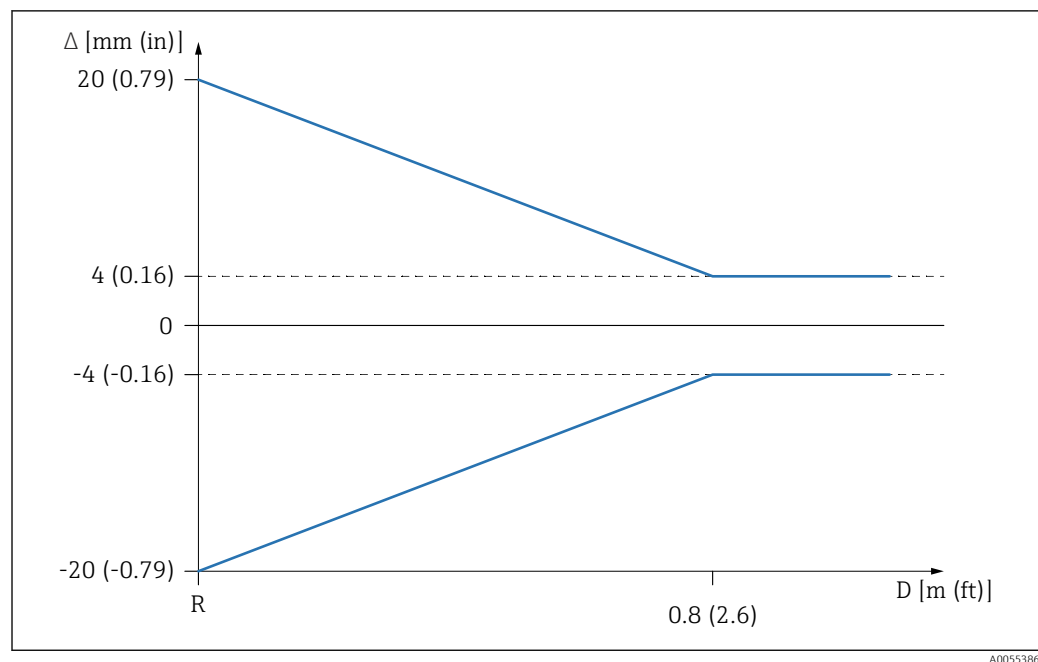
Differing values in near-range applications for liquids



 9 Maximum measurement error in near-range applications

- Δ Maximum measurement error
- R Reference point of the distance measurement
- D Distance from reference point of antenna

Differing values in near-range applications for solids



10 Maximum measurement error in near-range applications

Δ Maximum measurement error

R Reference point of the distance measurement

D Distance from reference point of antenna

Influence of ambient temperature

The output changes due to the effect of the ambient temperature with respect to the reference temperature.

The measurements are performed according to IEC 61298-3 / IEC 60770-1

Digital output (HART)

Average $T_C = \pm 2$ mm (± 0.08 in)/10 K

Analog (current output)

■ Zero point (4 mA): average $T_C = 0.02$ %/10 K

■ Span (20 mA): average $T_C = 0.05$ %/10 K

Reaction time

According to IEC 61298-2 / IEC 60770-1, the step response time is the time following an abrupt change in the input signal up until the changed output signal has adopted 90 % of the steady-state value for the first time.

The response time can be configured.

The following step response times apply (in accordance with IEC 61298-2/IEC 60770-1) when damping is switched off:

- Measuring rate ≤ 250 ms at operating voltage 24 V
- Step response time < 1 s

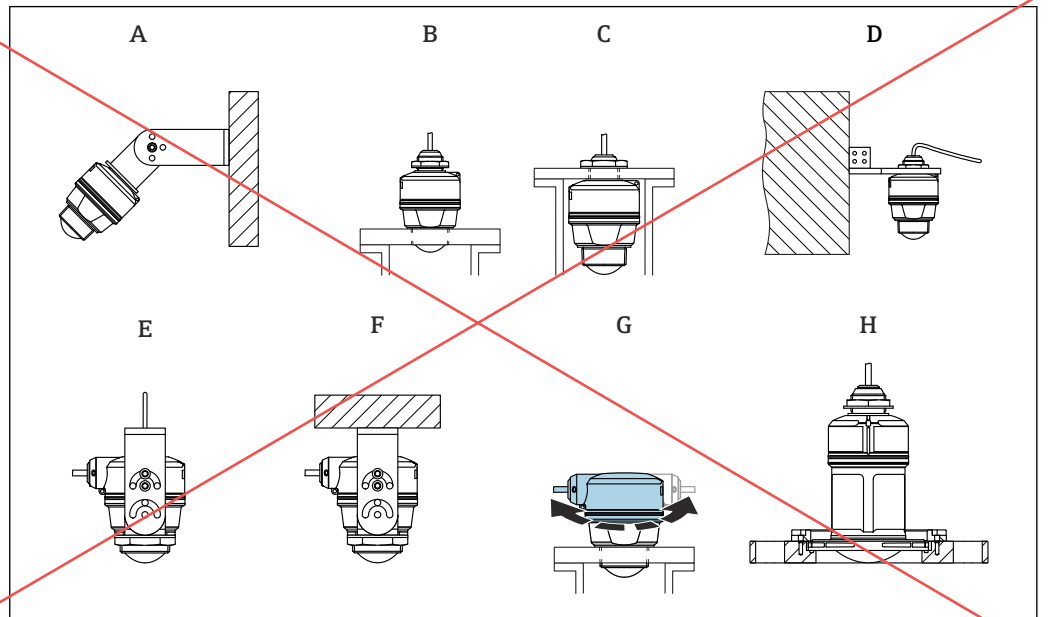
Warm-up time

The warm-up time (in accordance with IEC 62828-4) indicates the time required for the device to reach its maximum accuracy or performance after the supply voltage is energized.

Warm-up time: ≤ 15 s

Installation

Installation types



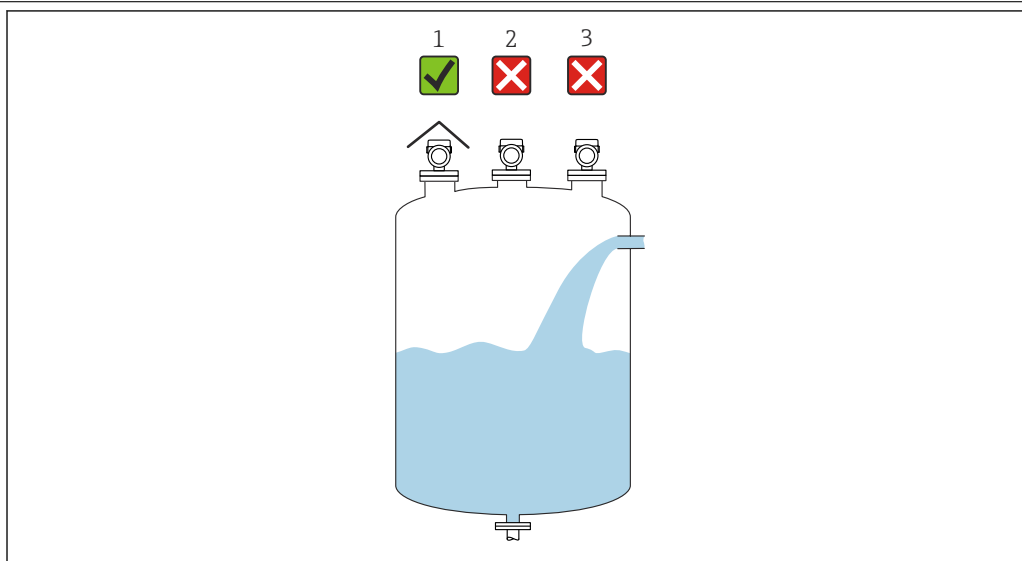
11 Wall or ceiling mount

- A Wall mounting adjustable
- B Tightened at antenna end process connection
- C Tightened at cable entry from above process connection
- D Wall mounting with cable entry from above process connection
- E Rope mounting with cable entry at the side
- F Ceiling mounting with cable entry at the side
- G Cable entry at the side, top housing section can be rotated
- H Mounting with UNI slip-on flange



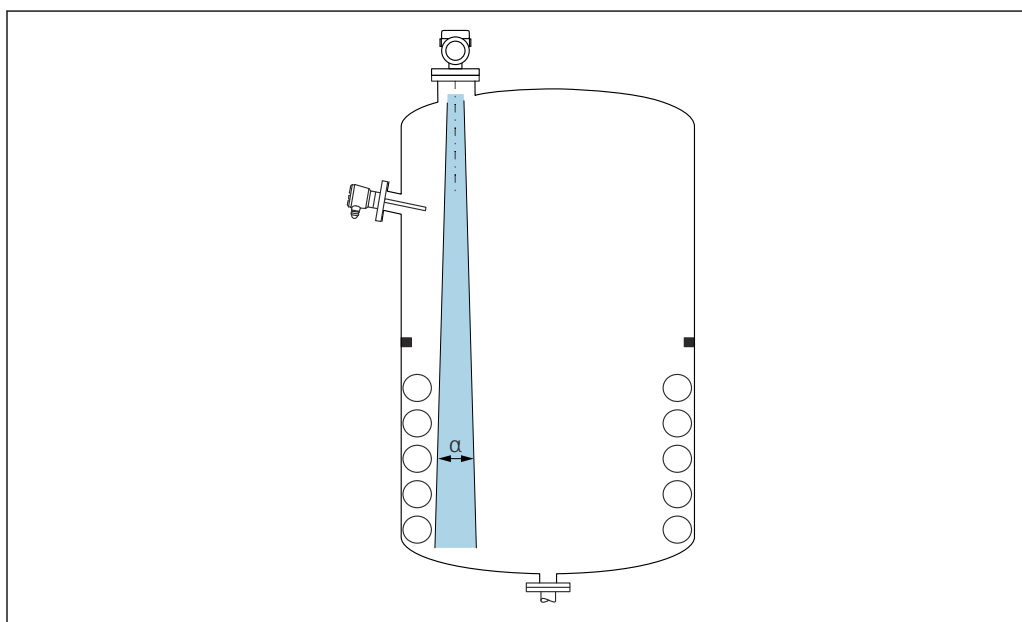
Please note the following:

- The sensor cables are not designed as supporting cables. Do not use them for suspension purposes.
- For rope mounting, the rope must be provided by the customer.
- Always operate the device in a vertical position in free-space applications.
- For devices with side cable outlet and 80 mm antenna, installation is only possible with a UNI slip-on flange.

Mounting location

A0055811

- 1 Use of a weather protection cover; protection from direct sunlight or rain
- 2 Installation not centered: Interferences can lead to incorrect signal analysis
- 3 Do not install above the filling curtain

Orientation**Internal vessel fittings**

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Avoid internal fittings (level switches, temperature sensors, struts, vacuum rings, heating coils, baffles etc.) inside the signal beam. Pay attention to the beam angle α .

Vertical alignment of antenna axis

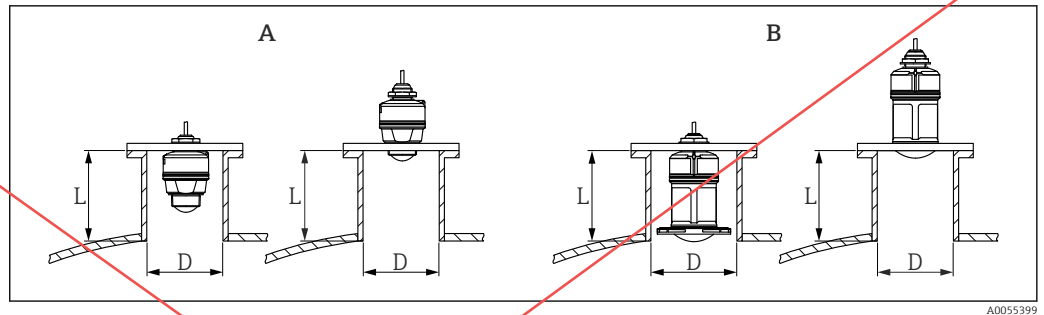
Align the antenna so that it is perpendicular to the product surface.



The maximum reach of the antenna can be reduced, or additional interference signals can occur, if the antenna is not installed perpendicular to the product.

Installation instructions

To ensure optimum measurement, the antenna must protrude from the nozzle. The interior of the nozzle must be smooth and must not contain any edges or welded joints. If possible, round the nozzle edge.



12 Nozzle installation

- A 40 mm (1.5 in) antenna
B 80 mm (3 in) antenna

The maximum nozzle length **L** depends on the nozzle diameter **D**.

Please note the limits for the diameter and length of the nozzle.

40 mm (1.5 in) antenna, installation outside nozzle

- D: min. 40 mm (1.5 in)
- L: max. $(D - 30 \text{ mm (1.2 in)}) \times 7.5$

40 mm (1.5 in) antenna, installation inside nozzle

- D: min. 80 mm (3 in)
- L: max. $100 \text{ mm (3.94 in)} + (D - 30 \text{ mm (1.2 in)}) \times 7.5$

80 mm (3 in) antenna, installation inside nozzle

- D: min. 120 mm (4.72 in)
- L: max. $140 \text{ mm (5.51 in)} + (D - 50 \text{ mm (2 in)}) \times 12$

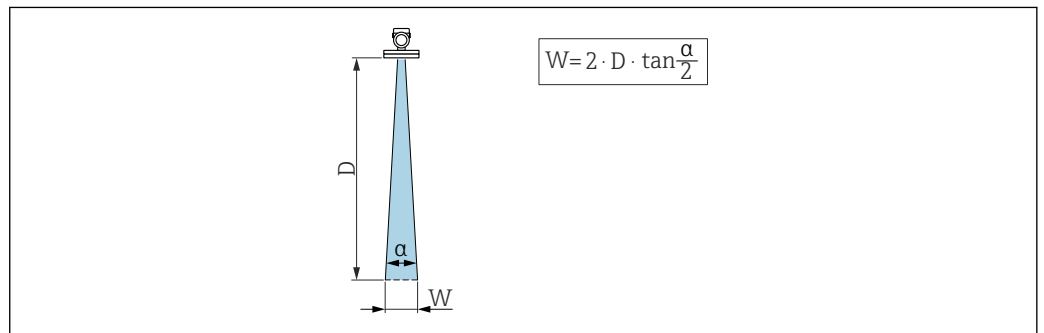
80 mm (3 in) antenna, installation outside nozzle

- D: min. 80 mm (3 in)
- L: max. $(D - 50 \text{ mm (2 in)}) \times 12$

Beam angle

Calculation

The beam angle is defined as the angle α at which the energy density of the radar waves reaches half the value of the maximum energy density (3dB width). Microwaves are also emitted outside the signal beam and can be reflected off interfering installations.

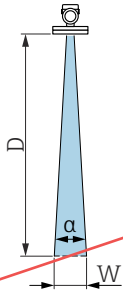


13 Relationship between beam angle α , distance **D** and beamwidth diameter **W**

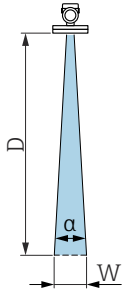


The beamwidth diameter **W** depends on the beam angle α and the distance **D**.

Antenna 40 mm (1.5 in), $\alpha = 8^\circ$

$W = D \times 0.14$	D	W
	5 m (16 ft)	0.70 m (2.29 ft)
	10 m (33 ft)	1.40 m (4.58 ft)
	15 m (49 ft)	2.09 m (6.87 ft)
	20 m (66 ft)	2.79 m (9.16 ft)

Antenna 80 mm (3 in), $\alpha = 4^\circ$

$W = D \times 0.07$	D	W
	5 m (16 ft)	0.35 m (1.15 ft)
	10 m (33 ft)	0.70 m (2.30 ft)
	15 m (49 ft)	1.05 m (3.45 ft)
	20 m (66 ft)	1.40 m (4.59 ft)
	25 m (82 ft)	1.75 m (5.74 ft)
	30 m (98 ft)	2.10 m (6.89 ft)

Special installation instructions**External measurement through plastic cover or dielectric windows**

- Dielectric constant of medium: $\epsilon_r \geq 10$
- The distance from the tip of the antenna to the tank should be approx. 100 mm (4 in).
- Avoid installation positions where condensate or buildup can form between the antenna and the vessel
- In the case of outdoor installations, ensure that the area between the antenna and the tank is protected from the weather
- Do not install any fittings or attachments between the antenna and the tank that could reflect the signal

The thickness of the tank ceiling or the dielectric window depends on the ϵ_r of the material.

The material thickness can be a full multiple of the optimum thickness (table); it is important to note, however, that the microwave transparency decreases significantly with increasing material thickness.

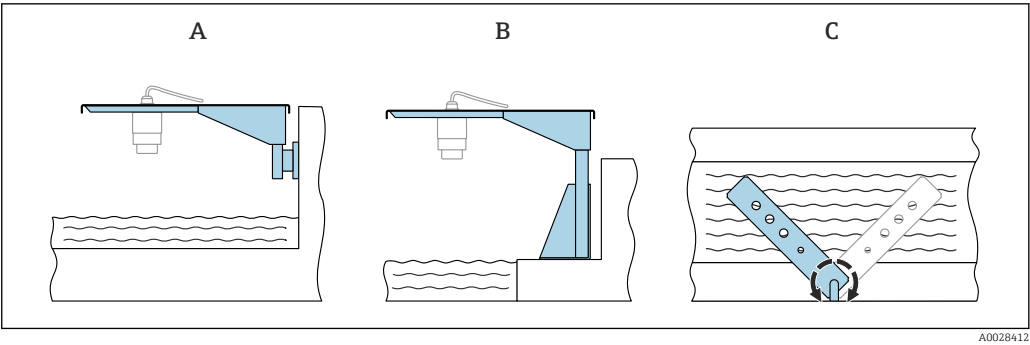
Optimum material thickness

Material	Optimum material thickness
PE; ϵ_r 2.3	1.25 mm (0.049 in)
PTFE; ϵ_r 2.1	1.30 mm (0.051 in)
PP; ϵ_r 2.3	1.25 mm (0.049 in)
Perspex; ϵ_r 3.1	1.10 mm (0.043 in)

Weather protection cover

A weather protective cover is recommended for outdoor use.

The weather protective cover can be ordered as an accessory or together with the device via the product structure "Accessory enclosed".



18 Cantilever installation, with pivot

- A Cantilever with wall bracket (side view)
- B Cantilever with mounting frame (side view)
- C Cantilever can be turned, e.g. in order to position the device over the center of the flume (top view)

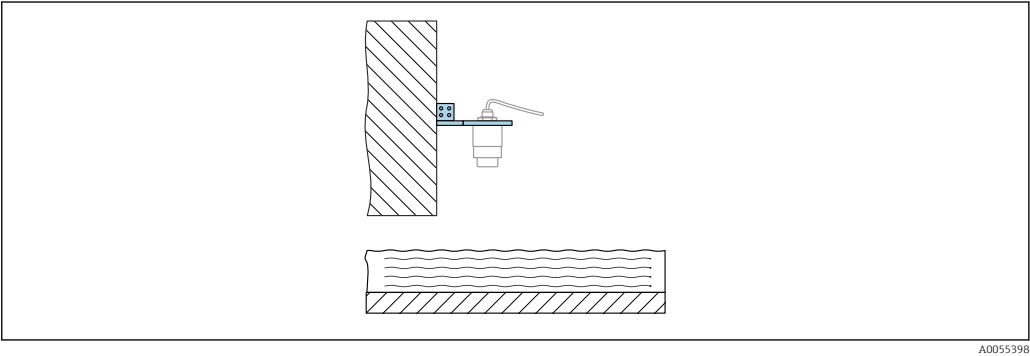
NOTICE

There is no conductive connection between the mounting bracket and transmitter housing.
Electrostatic charging possible.

- Integrate the mounting bracket in the local potential equalization system.

Mounting with a pivotable mounting bracket

The pivotable mounting bracket can be ordered as an accessory or together with the device via the product structure "Accessory enclosed".



19 Pivotable and adjustable cantilever with wall bracket (e.g. to align the device with the center of a flume)

NOTICE

There is no conductive connection between the mounting bracket and transmitter housing.
Electrostatic charging possible.

- Integrate the mounting bracket in the local potential equalization system.

Environment

Ambient temperature range	-40 to +80 °C (-40 to +176 °F) If operating outdoors in strong sunlight: ■ Mount the device in the shade. ■ Avoid direct sunlight, particularly in warmer climatic regions. ■ Use a weather protective cover.
Storage temperature	-40 to +80 °C (-40 to +176 °F)
Climate class	According to IEC 60068-2-38 test Z/AD (relative humidity 4 to 100 %).
Operating height	Up to 5 000 m (16 404 ft) above sea level

Degree of protection	Testing according to IEC 60529 and NEMA 250: ■ IP66, NEMA Type 4X ■ IP68, NEMA Type 6P (24 h at 1.83 m (6.00 ft) under water)
Vibration resistance	■ Stochastic noise (random sweep) as per IEC 60068-2-64 Case 2 ■ Guaranteed for 5 to 2 000 Hz: $1.25 \text{ (m/s}^2\text{)}^2\text{/Hz}$, ~ 5 g
Electromagnetic compatibility (EMC)	■ Electromagnetic compatibility as per EN 61326 series and NAMUR recommendation EMC (NE21) ■ Maximum measured error during EMC testing: < 0.5 % of the span. For more details, refer to the EU Declaration of Conformity (www.endress.com/downloads).

Process

Process temperature, process pressure



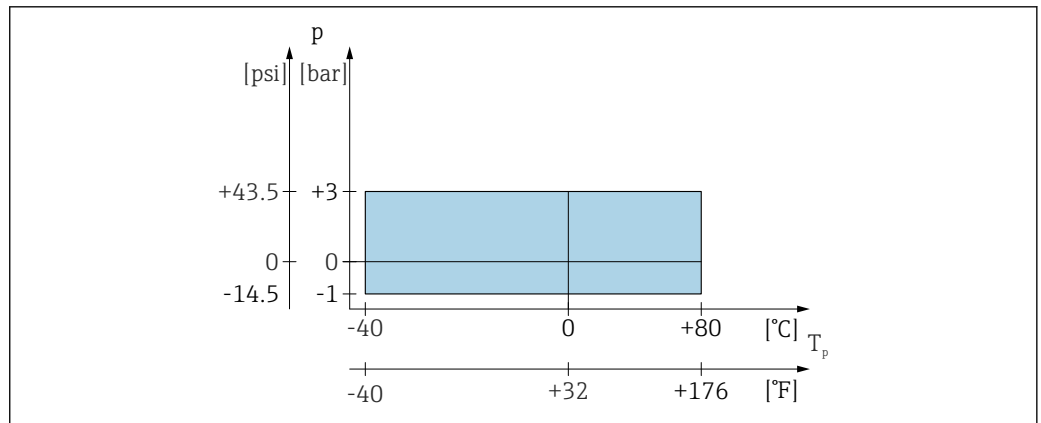
The maximum pressure for the device depends on the lowest-rated element with regard to pressure.

Components are: process connection, optional mounting parts, or accessories.

WARNING

Incorrect design or use of the device may cause injury due to bursting parts!

- ▶ Only operate the device within the specified limits for the components!
- ▶ MWP (Maximum Working Pressure): The MWP is specified on the nameplate. This value refers to a reference temperature of +20 °C (+68 °F) and may be applied to the device for an unlimited time. Observe the temperature dependency of the maximum working pressure. For flanges, refer to the following standards for the permitted pressure values at higher temperatures: EN 1092-1 (with regard to their stability/temperature property, the materials 1.4435 and 1.4404 are grouped together under EN 1092-1; the chemical composition of the two materials can be identical), ASME B16.5, JIS B2220 (the latest version of the standard applies in each case). Maximum working pressure data that deviate from this are provided in the relevant sections of the Technical Information.
- ▶ The Pressure Equipment Directive (2014/68/EU) uses the abbreviation **PS**. This corresponds to the maximum working pressure (MWP) of the device.



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20 Permitted range for process temperature and process pressure

Process temperature range

–40 to +80 °C (–40 to +176 °F)

Process pressure range, 40 mm (1.5 in) antenna

- $p_{\text{gauge}} = -1 \text{ to } 3 \text{ bar}$ (–14.5 to 43.5 psi)
- $p_{\text{abs}} < 4 \text{ bar}$ (58 psi)

Process pressure range, 80 mm (3 in) antenna with UNI slip-on flange 3", 4"

- $p_{\text{gauge}} = -1 \text{ to } 1 \text{ bar}$ (–14.5 to 14.5 psi)
- $p_{\text{abs}} < 2 \text{ bar}$ (29 psi)

Process pressure range, 80 mm (3 in) antenna with UNI slip-on flange 6"

For unpressurized applications



The pressure range may be further restricted in the case of a CRN approval.

Relative permittivity

For liquids

- $\epsilon_r \geq 1.8$
- Contact Endress+Hauser for lower ϵ_r values

For bulk solids

$\epsilon_r \geq 1.6$

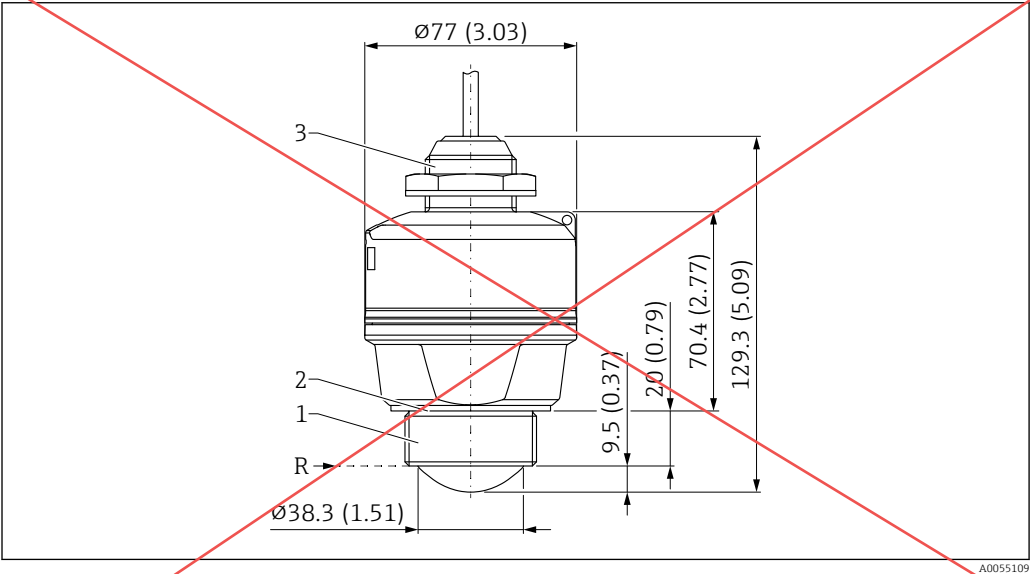
For applications with a lower relative permittivity than indicated, contact Endress+Hauser.

-  For the relative permittivity values (ϵ_r values) of many media commonly used in industry, please refer to:
- Relative permittivity (ϵ_r value), Compendium CP01076F
 - The Endress+Hauser "DC Values app" (available for Android and iOS)

Mechanical construction

Dimensions

40 mm (1.5 in) antenna, cable entry from above

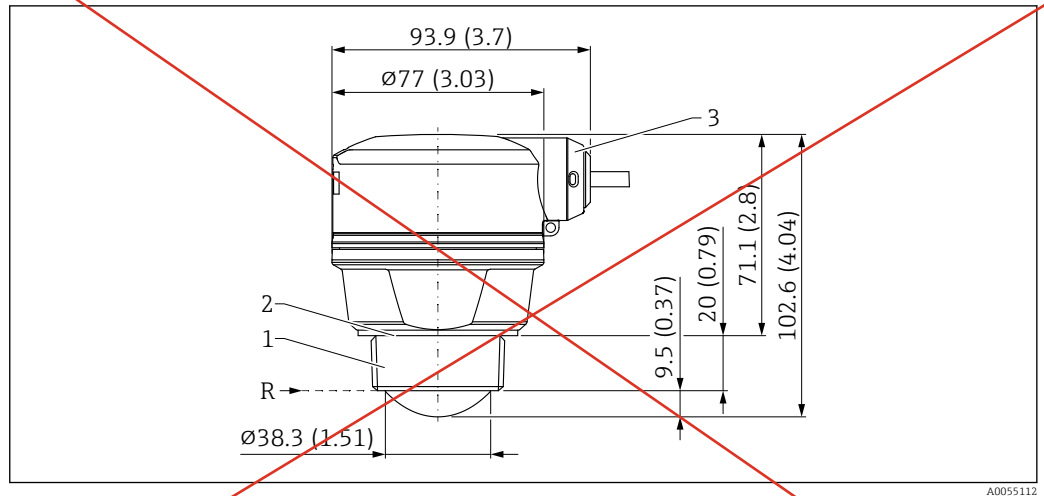


 21 Dimensions; 40 mm (1.5 in) antenna with cable entry from above. Unit of measurement mm (in)

- R Reference point of the measurement
1 Antenna end process connection, thread
2 EPDM seal (G 1 1/2 thread)
3 Cable entry from above process connection

 The seal thickness is 2 mm (0.08 in).

40 mm (1.5 in) antenna, cable entry at the side

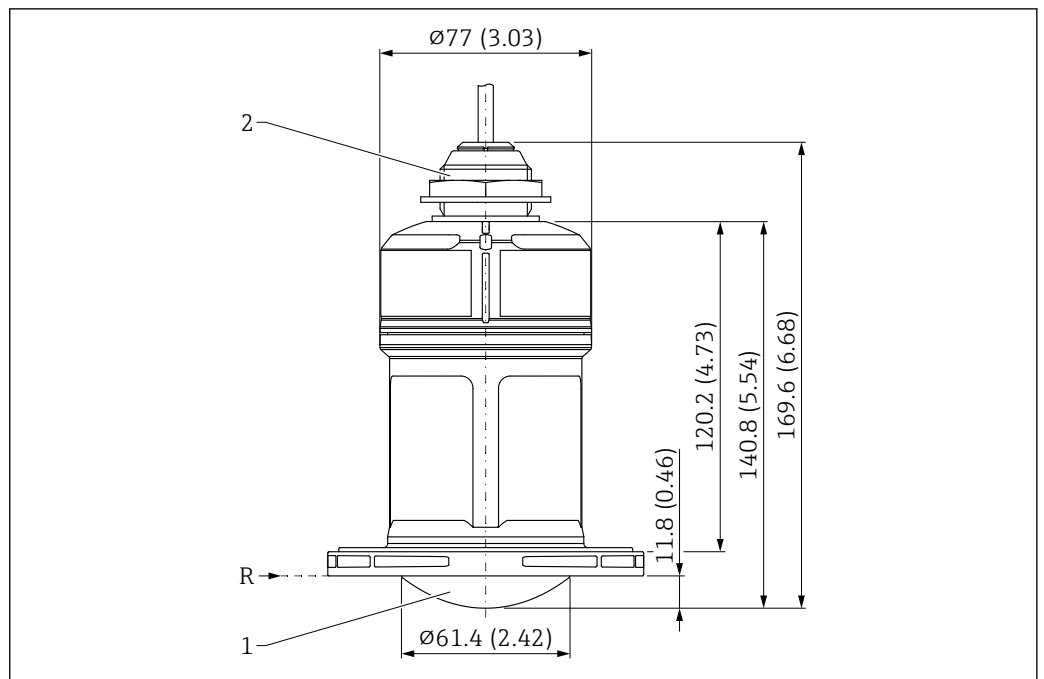


22 Dimensions; 40 mm (1.5 in) antenna with cable entry at the side. Unit of measurement mm (in)

- R Reference point of the measurement
 1 Antenna end process connection, thread
 2 EPDM seal (G 1 1/2 thread)
 3 Side cable entry

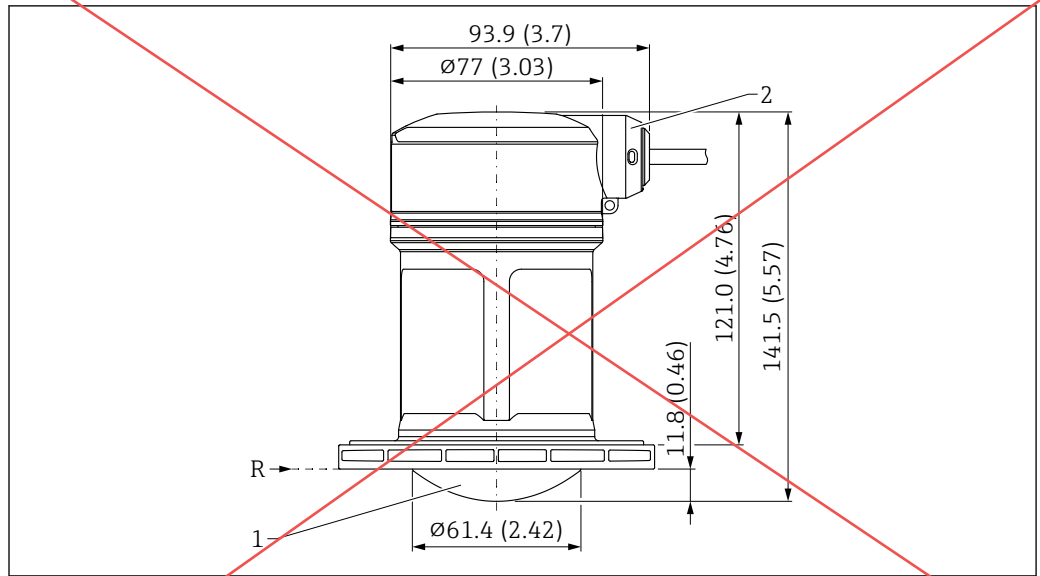
i The seal thickness is 2 mm (0.08 in).

80 mm (3 in) antenna, cable entry from above



23 Dimensions; 80 mm (3 in) antenna with cable entry from above. Unit of measurement mm (in)

- R Reference point of the measurement
 1 Antenna end process connection, without; prepared for UNI slip-on flange
 2 Cable entry from above process connection

80 mm (3 in) antenna, cable entry at the side

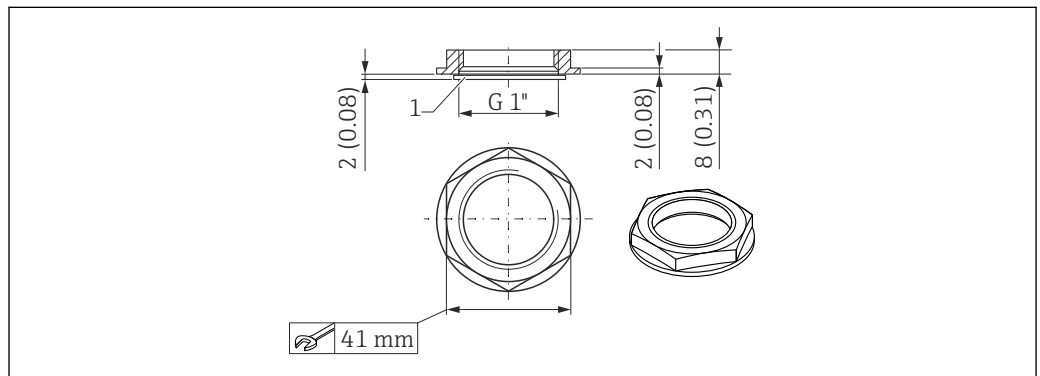
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24 Dimensions; 80 mm (3 in) antenna with cable entry at the side

R Reference point of the measurement

1 Antenna end process connection, without; prepared for UNI slip-on flange

2 Side cable entry

Counter nut, cable entry from above process connection

A0028419

25 Dimensions; counter nut, cable entry from above process connection. Unit of measurement mm (in)

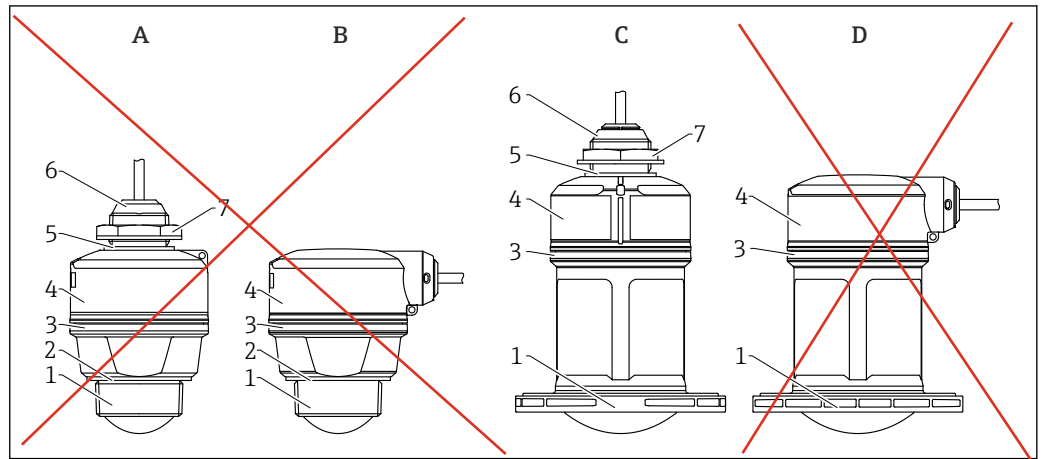
1 Seal

- The counter nut with seal (EPDM) is included in the scope of delivery
- Material: PA6.6

Weight**Weight (including 5 m (16 ft) cable)**

- Device with 40 mm (1.5 in) antenna: approx. 0.5 kg (1.1 lb)
- Device with 80 mm (3 in) antenna: approx. 0.7 kg (1.5 lb)

Materials



26 Device design

~~A 40 mm (1.5 in) antenna, cable entry from above~~

~~B 40 mm (1.5 in) antenna, cable entry at the side~~

~~C 80 mm (3 in) antenna, cable entry from above~~

~~D 80 mm (3 in) antenna, cable entry at the side~~

1 Antenna end process connection; PVDF

2 EPDM seal (for G 1½" thread)

3 PBT/PC design ring

4 Sensor housing/Cable entry process connection; PBT/PC (for dust ignition-proof devices: PC)

5 EPDM seal

6 Cable entry; PBT/PC (for dust ignition-proof devices: PC)

7 Counter nut; PA6.6

Connecting cable

Available cable length: 5 to 300 m (16 to 980 ft)

Material : PVC

For devices with marine approval: halogen-free cable (material: XLPE = connected polyethylene)

Operability

Operating concept

Operator-oriented menu structure for user-specific tasks

- Guidance
- Diagnostics
- Application
- System

Fast and safe commissioning

- Interactive wizard with graphical interface for guided commissioning in FieldCare/DeviceCare or SmartBlue app
- Menu guidance with brief descriptions of the individual parameter functions

Integrated data memory

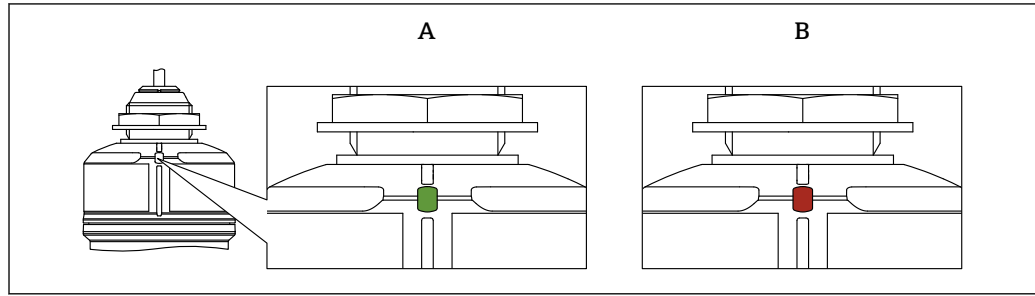
Up to 100 event messages recorded in the device

Efficient diagnostic behavior increases measurement reliability

- Remedial action is integrated in plain text
- Diverse simulation options

Bluetooth® wireless technology (optional)

- Quick and easy setup with the SmartBlue app or Field Xpert SMT70/SMT77
- No additional tools or adapters required
- Encrypted single point-to-point data transmission (tested by Fraunhofer Institute) and password-protected communication via Bluetooth® wireless technology
- The device can be retrofitted with Bluetooth® wireless technology

LED indicator

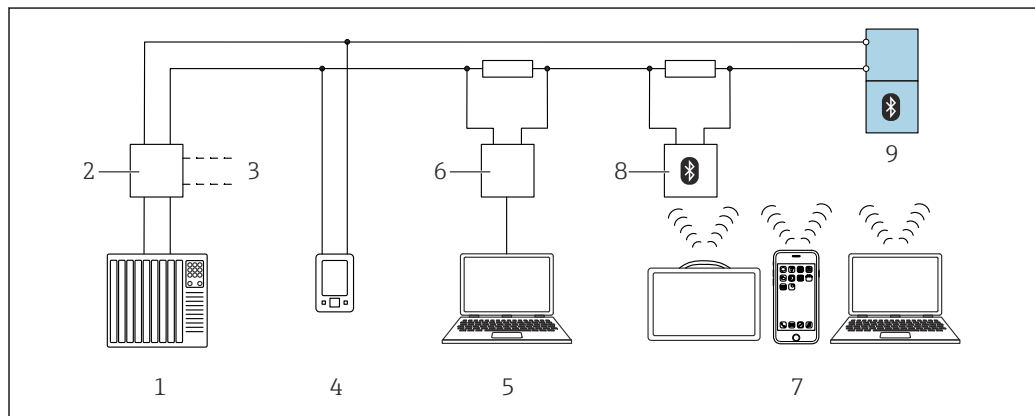
A0055146

27 LED indicator on the device

- A Green LED
B Red LED

Functions:

- Displays the operating status
 - Operation (green)
 - Fault (red)
- Displays an active Bluetooth® connection (flashing)

Remote operation**Via HART protocol or Bluetooth® wireless technology**

A0044334

28 Options for remote operation via HART protocol

- 1 PLC (programmable logic controller)
- 2 Transmitter power supply unit, e.g. RN42 (with communication resistor)
- 3 Connection for Commubox FXA195 and AMS Trex™ device communicator
- 4 AMS Trex™ device communicator
- 5 Computer with operating tool (e.g. DeviceCare/FieldCare, AMS Device View, SIMATIC PDM)
- 6 Commubox FXA195 (USB)
- 7 Field Xpert SMT70/SMT77, smartphone or computer with operating tool (e.g. DeviceCare)
- 8 Bluetooth® modem with connecting cable (e.g. VIATOR)
- 9 Transmitter

Operation via Bluetooth® wireless technology (optional)**Prerequisite**

- Device with Bluetooth® wireless technology order option
- Smartphone or tablet with Endress+Hauser SmartBlue app or PC with DeviceCare from version 1.07.07 or Field Xpert SMT70/SMT77

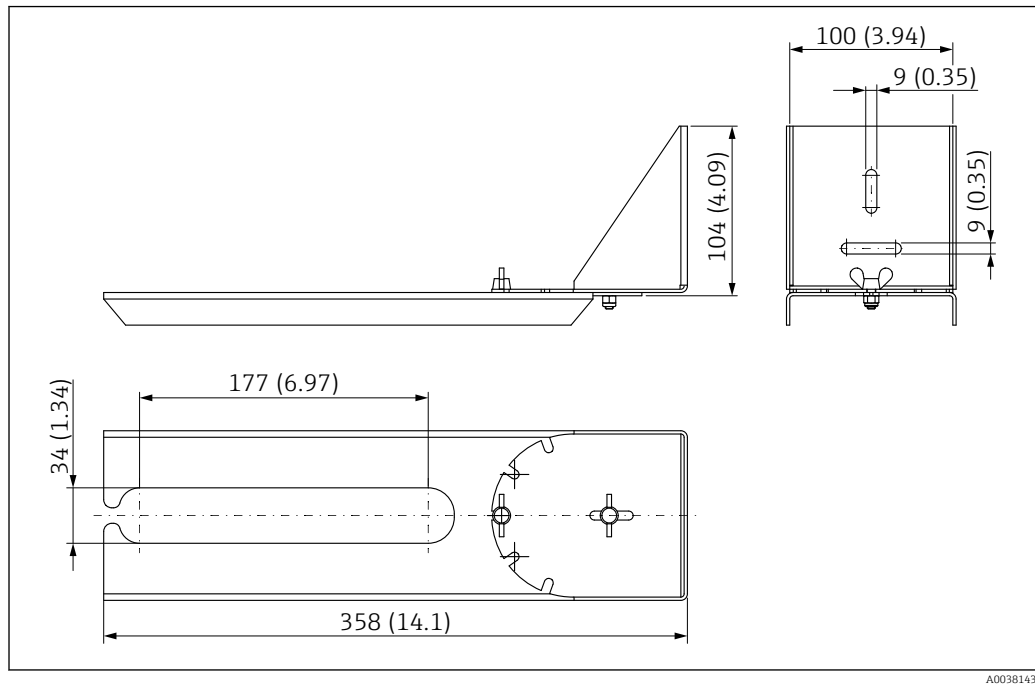
The connection has a range of up to 25 m (82 ft). The range can vary depending on environmental conditions such as attachments, walls or ceilings.

Supported operating tools

Smartphone or tablet with Endress+Hauser SmartBlue app, DeviceCare from version 1.07.07, FieldCare, AMS and PDM

Pivotable mounting bracket

The pivotable mounting holder is used, for example, to install the device in a manhole over a sewer channel.



51 Dimensions of pivotable mounting bracket. Unit of measurement mm (in)

i 34 mm (1.34 in) openings for all G 1" or MNPT 1" cable entry thread process connections

Material

316L (1.4404)

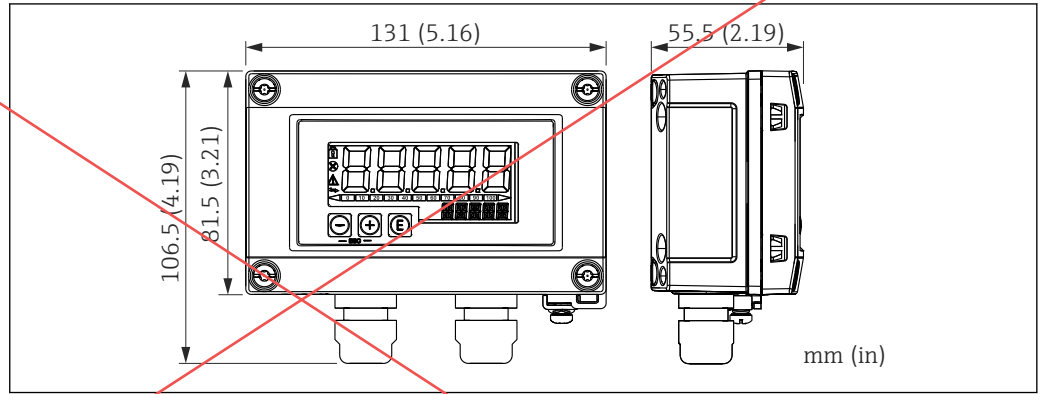
Order code

71429910

FAU40 alignment unit

The alignment unit is used to optimally align the sensor with the bulk solids.

RIA15 in the field housing



A0017722

60 Dimensions of RIA15 in the field housing. Unit of measurement mm (in)

i Older device versions before February 2025 can only be connected via the 4 to 20 mA current output.

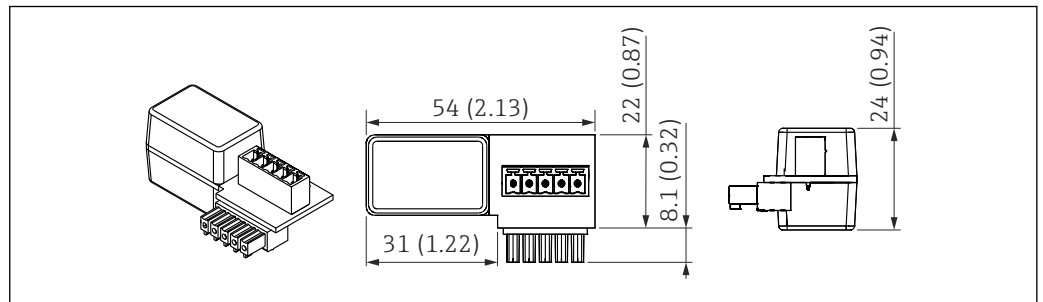
i The remote display RIA15 can be ordered with or without operation via the product structure "Accessory enclosed".

Field housing material: Plastic (PBT with steel fibers, antistatic)

Other housing versions are available via the RIA15 product structure.

i Alternatively available as an accessory, for details see Technical Information TI01043K and Operating Instructions BA01170K

HART communication resistor

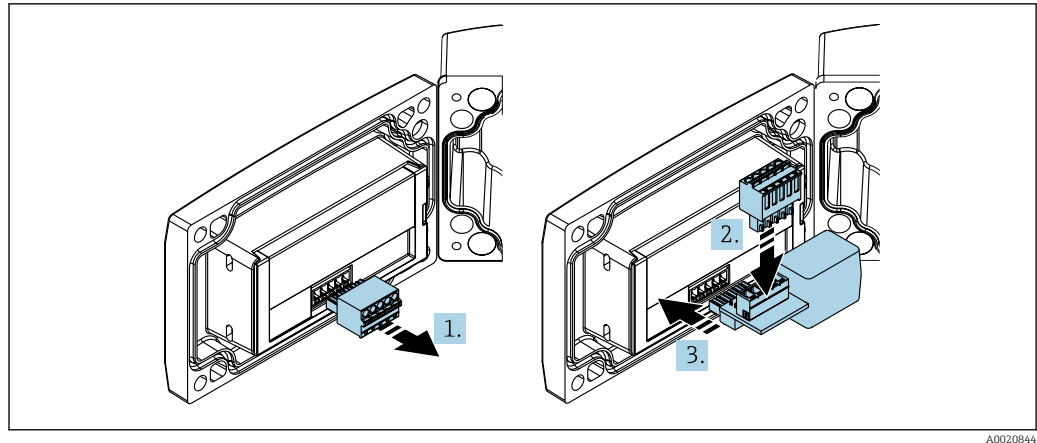


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61 Dimensions of HART communication resistor. Unit of measurement mm (in)

The HART communication resistor is required for operation of the RIA15 and is supplied with "remote display RIA15, with operation via HART" when ordering.

i Document Technical Information TI01043K and Operating Instructions BA01170K



A0020844

i In order to operate the RIA15, the HART communication resistor must be integrated.

1. Disconnect the plug-in terminal block.
2. Insert the terminal block into the slot provided on the HART communication resistor module.
3. Insert the HART communication resistor in the slot in the housing.

DeviceCare SFE100

Configuration tool for IO-Link, HART, PROFIBUS and FOUNDATION Fieldbusfield devices
DeviceCare is available for download free of charge at www.software-products.endress.com. You need to register in the Endress+Hauser software portal to download the application.

 Technical Information TI01134S

FieldCare SFE500

FDT-based plant asset management tool
It can configure all intelligent field units in your system and helps you manage them. By using the status information, it is also a simple but effective way of checking their status and condition.

 Technical Information TI00028S

Device Viewer

All the spare parts for the device, along with the order code, are listed in the *Device Viewer* (www.endress.com/deviceviewer).

Commubox FXA195 HART

For intrinsically safe HART communication with FieldCare via the USB interface

 Technical Information TI00404F

RN22

Single or two-channel active barrier for safe electrical isolation of 4 to 20 mA standard signal circuits, HART transparent

 Technical Information TI01515K and Operating Instructions BA02004K

RN42

Single-channel active barrier with wide-range power supply for safe electrical isolation of 4 to 20 mA standard signal circuits, HART transparent.

 Technical Information TI01584K and Operating Instructions BA02090K

Field Xpert SMT70

Universal, high-performance tablet PC for device configuration in Ex Zone 2 and non-Ex areas

 For details, see "Technical Information" TI01342S

Field Xpert SMT77

Universal, high-performance tablet PC for device configuration in Ex Zone 1 areas

 For details, see "Technical Information" TI01418S

SmartBlue app	Mobile app for easy configuration of devices on site via Bluetooth® wireless technology.
RMA42	Digital process transmitter for monitoring and displaying analog measured values  For details, see Technical Information TI00150R and Operating Instructions BA00287R

Documentation



For an overview of the scope of the associated Technical Documentation, refer to the following:

- *Device Viewer* (www.endress.com/deviceviewer): Enter the serial number from the nameplate
- *Endress+Hauser Operations app*: Enter serial number from nameplate or scan matrix code on nameplate.

Standard documentation	Document type: Operating Instructions (BA) Installation and initial commissioning – contains all the functions in the operating menu that are needed for a routine measuring task. Functions beyond this scope are not included. Document type: Description of Device Parameters (GP) The document is part of the Operating Instructions and serves as a reference for parameters, providing a detailed explanation of each individual parameter of the operating menu. Document type: Brief Operating Instructions (KA) Quick guide to the first measured value – includes all essential information from incoming acceptance to electrical connection. Document type: Safety Instructions, certificates Depending on the approval, safety instructions are supplied with the device, e.g. XA. This documentation is an integral part of the Operating Instructions. Information on the Safety Instructions (XA) that are relevant for the device is provided on the nameplate.
Supplementary device-dependent documentation	Additional documents are supplied depending on the device version ordered: Always comply strictly with the instructions in the supplementary documentation. The supplementary documentation is an integral part of the device documentation.

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